Object-Oriented Software Design - Questions & Answers

1 Mark Questions

1. What does OOP stand for?

- OOP stands for Object-Oriented Programming.

2. What is an object in OOP?

- An object is an entity that has states and behaviors.

3. What are the four building blocks of OOP?

- Inheritance, Encapsulation, Abstraction, Polymorphism.

4. What is the IS-A relationship in OOP?

- It is an inheritance relationship where a child class inherits from a parent class.

5. How is encapsulation achieved in Java?

- By declaring variables as private and providing public getter and setter methods.

2 Mark Questions

1. Define a class in OOP.

- A class is a blueprint or template from which objects are created. It defines the properties and behaviors of objects.

2. What is procedural programming?

- Procedural programming is a programming paradigm that follows a sequential, step-by-step approach using procedures or functions.

3. Explain abstraction with an example.

- Abstraction hides the implementation details and only shows functionality. Example: A car's accelerator increases speed without revealing how it works.

4. Differentiate between method overloading and method overriding.

- Method overloading occurs when multiple methods have the same name but different

parameters. Method overriding occurs when a child class redefines a method from its parent class.

3 Mark Questions

1. Explain the concept of inheritance with an example.

- Inheritance allows a child class to acquire the properties and behaviors of a parent class.

Example: A "Dog" class can be a parent, while "Pomeranian" and "Bulldog" are child classes.

2. What are the advantages of OOP over procedural programming?

- OOP promotes code reusability, better data security, scalability, and easier maintenance through encapsulation and inheritance.

3. How does polymorphism enhance flexibility in OOP?

- Polymorphism allows objects to be treated as instances of their parent class, enabling dynamic method dispatch and flexibility in method implementation.

5 Mark Questions

1. Discuss the differences between Procedural Programming and Object-Oriented Programming.

- Procedural Programming:
  - Follows a step-by-step sequential execution.
  - Uses functions and procedures for code reuse.
  - Less emphasis on data security.
- Object-Oriented Programming:
  - Organizes code into objects and classes.
  - Uses encapsulation, inheritance, and polymorphism.
  - Provides better data security and reusability.

2. Describe encapsulation in detail and how it is implemented in Java.

- Encapsulation is the process of wrapping data and code into a single unit.
- It protects data from unauthorized access by making variables private and using public methods to access them.

- In Java, encapsulation is achieved by:
  - Declaring class variables as private.
  - Providing public getter and setter methods.
  - Example:

```
class Student {
    private String name;

    public void setName(String newName) {
        name = newName;
    }

    public String getName() {
        return name;
    }
}
```